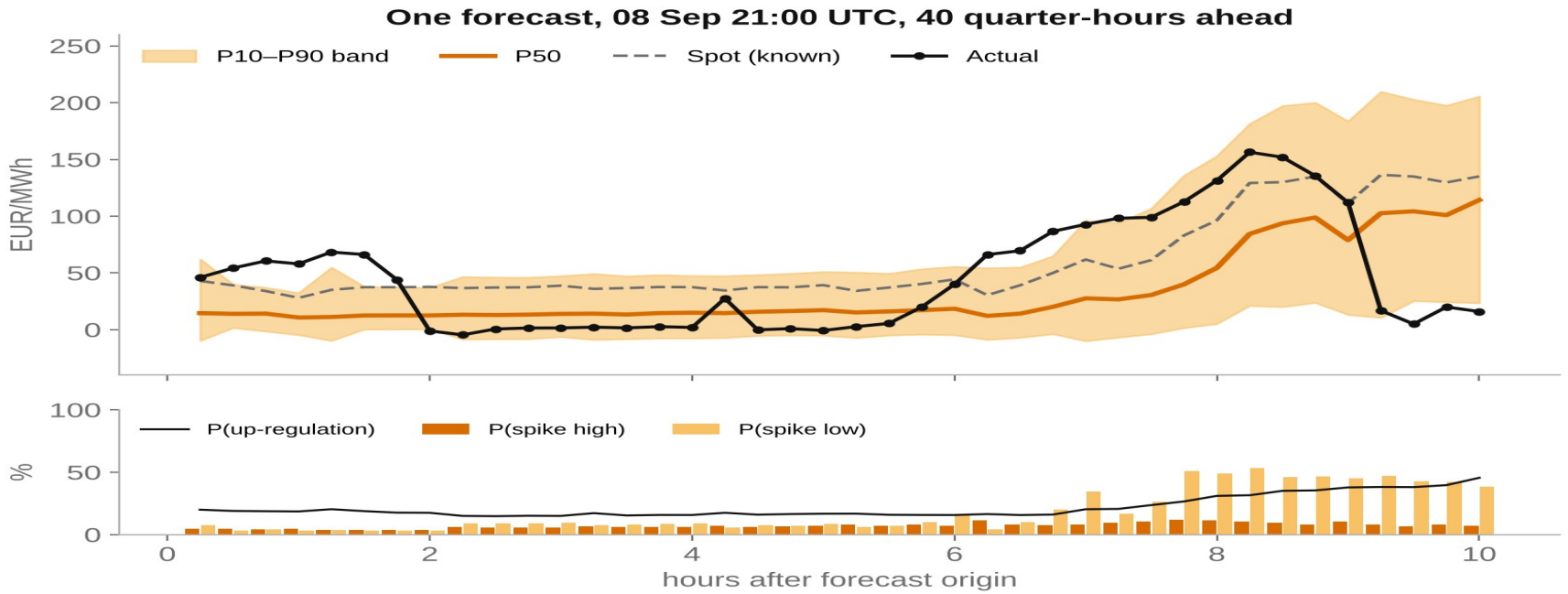
THE RESULT

A 10-hour forecast with a calibrated band and spike risk



–38 %

P50 error

MAE 66 → 41 EUR/MWh

–53 %

band error

pinball 24 → 11

82 %

band coverage

target 80 %, baseline 68 %

–23 %

spike Brier

0.113 → 0.087

–15 %

direction Brier

0.451 → 0.385

Validation 7 Sep – 4 Oct 2026 vs. the organiser's baseline. Better in 100 % of day-block bootstrap resamples on all four metrics.

The price is mechanical and the error lives in the spikes

Weather
wind, temperature

→

Plan error
system imbalance

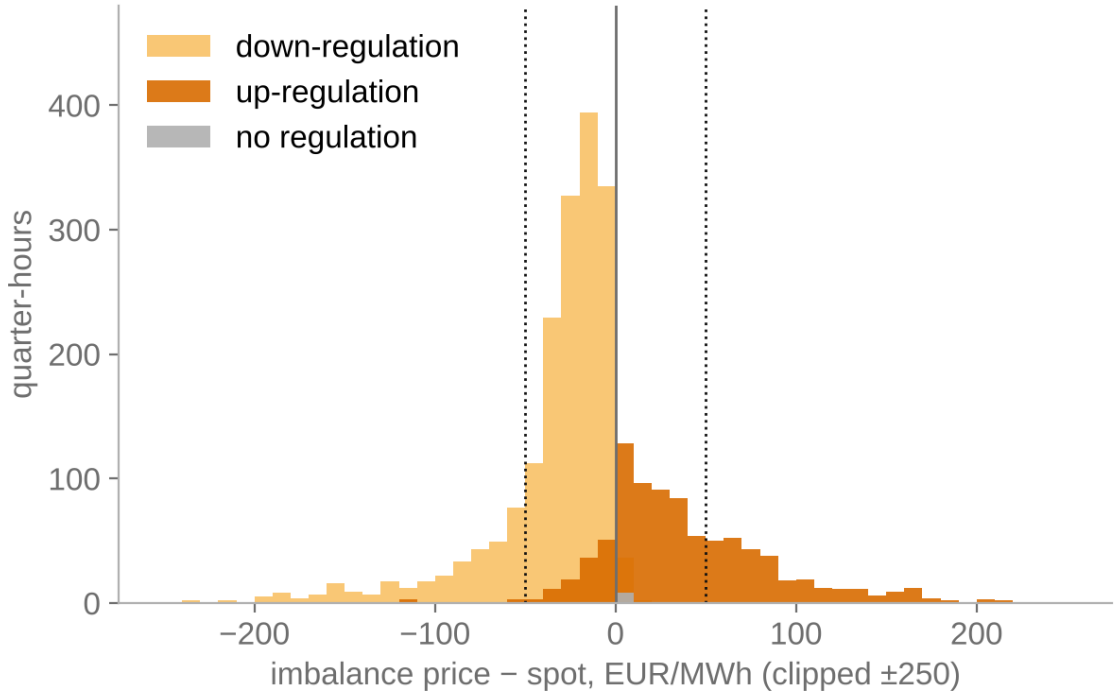
→

Regulation: mFRR + aFRR
mFRR sets the direction, both set the price

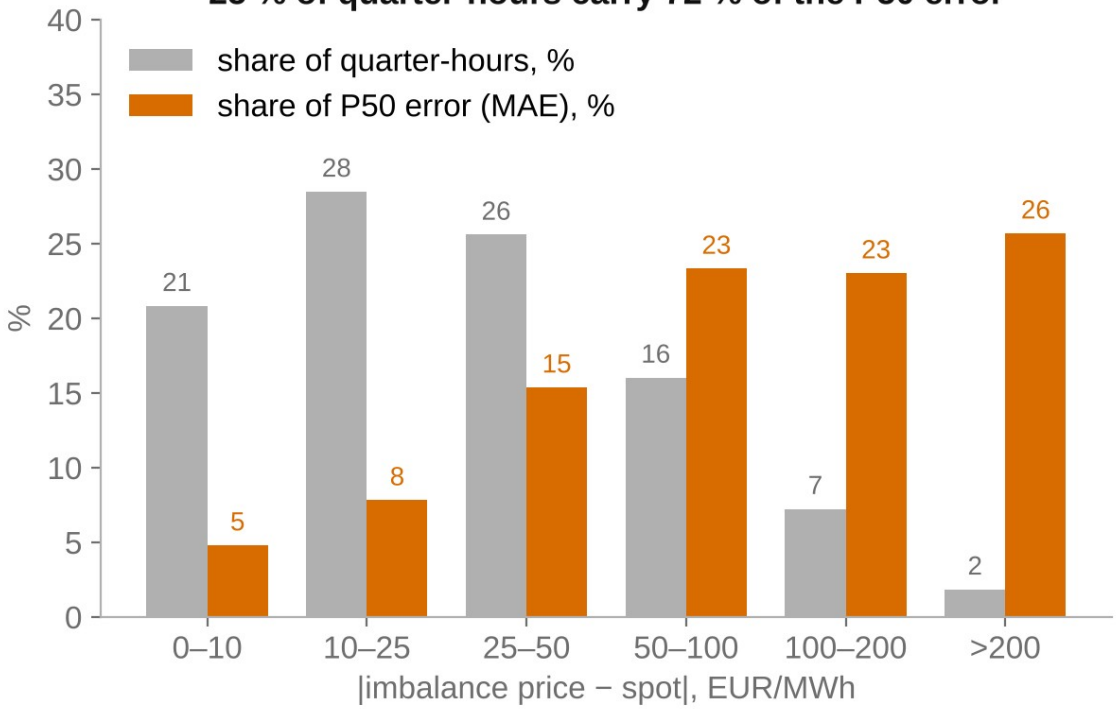
→

Imbalance price
Fingrid rule, reproduced 99.6 %

The spread is bimodal, the regulation direction picks the side

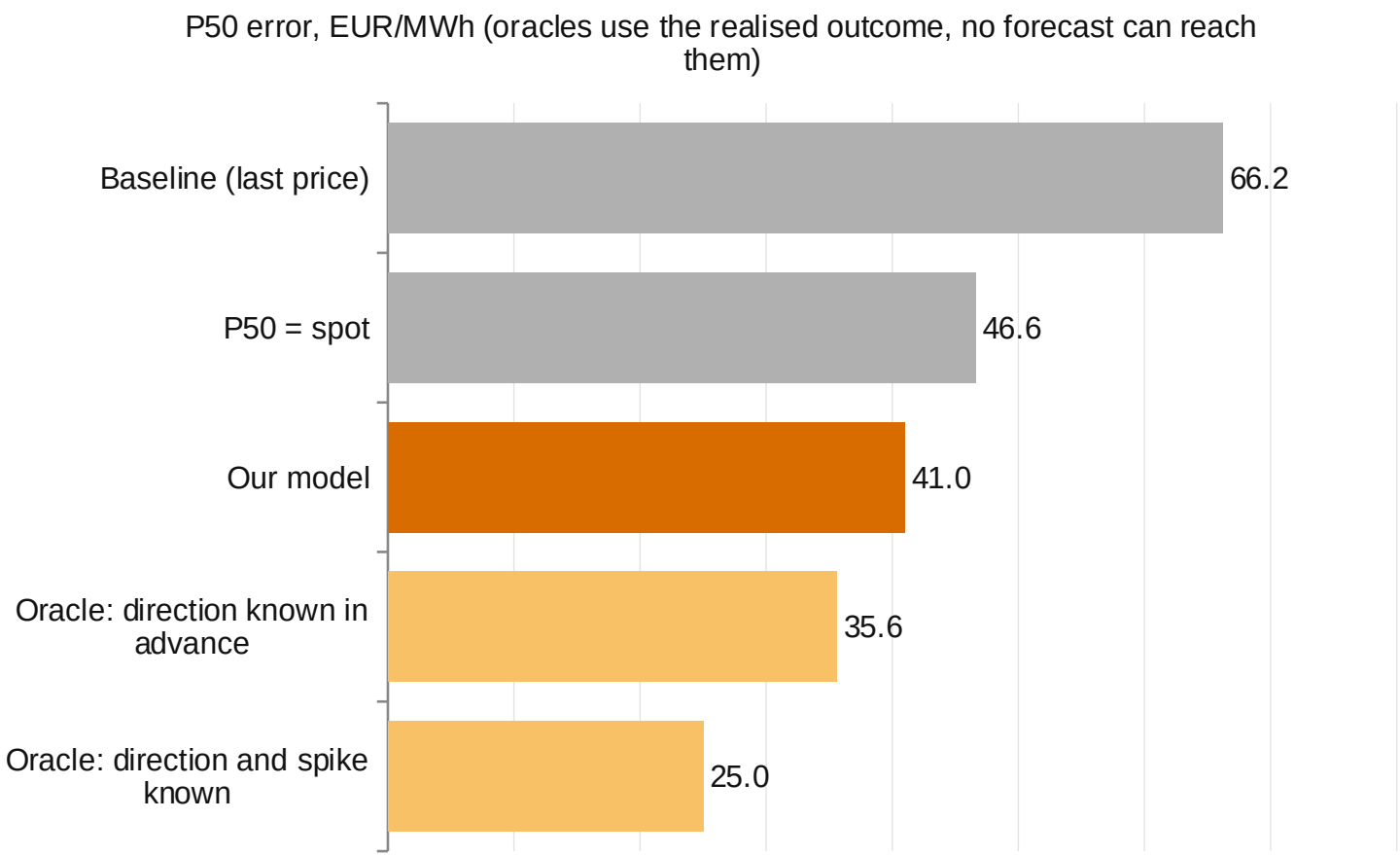


25 % of quarter-hours carry 72 % of the P50 error



Spot is known. We forecast only the spread: first the direction, then the price given the direction.

The data sets a floor that no forecast of this price can pass

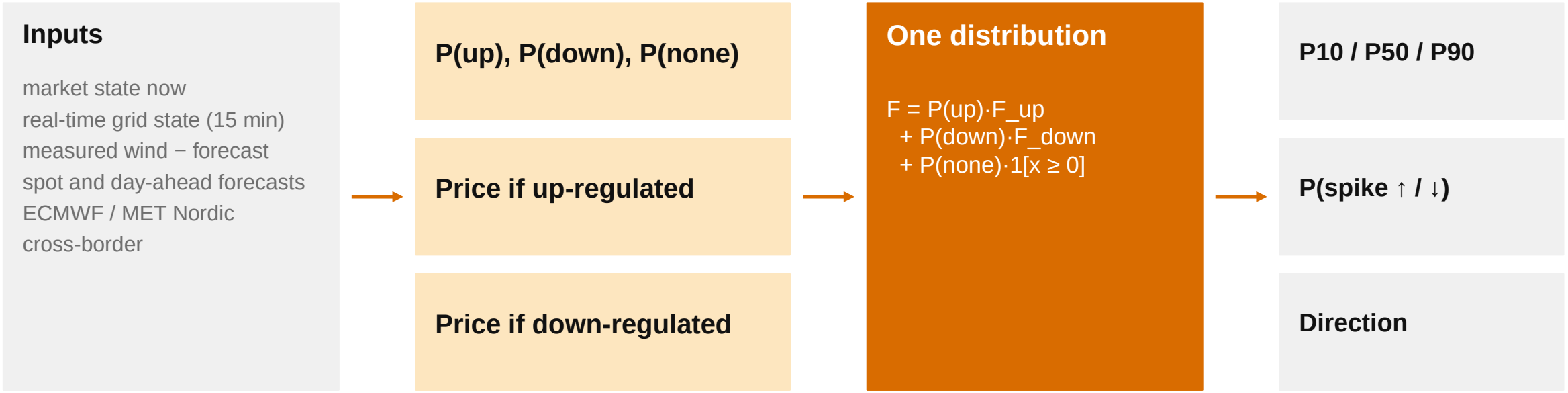


What the data told us

- Spot alone already gets to 46.6: the game is the spread, not the level
- Knowing the direction perfectly 10 h ahead would still leave 35.6: the rest is spike timing
- Shrinking P50 toward spot (0.5×) hurts: 43.2. The point forecast is already where the data wants it
- Published 15-min models beat persistence by 12–25 %; we beat it by 38 %

So the value is in calibrated probabilities and the band, not in polishing P50.

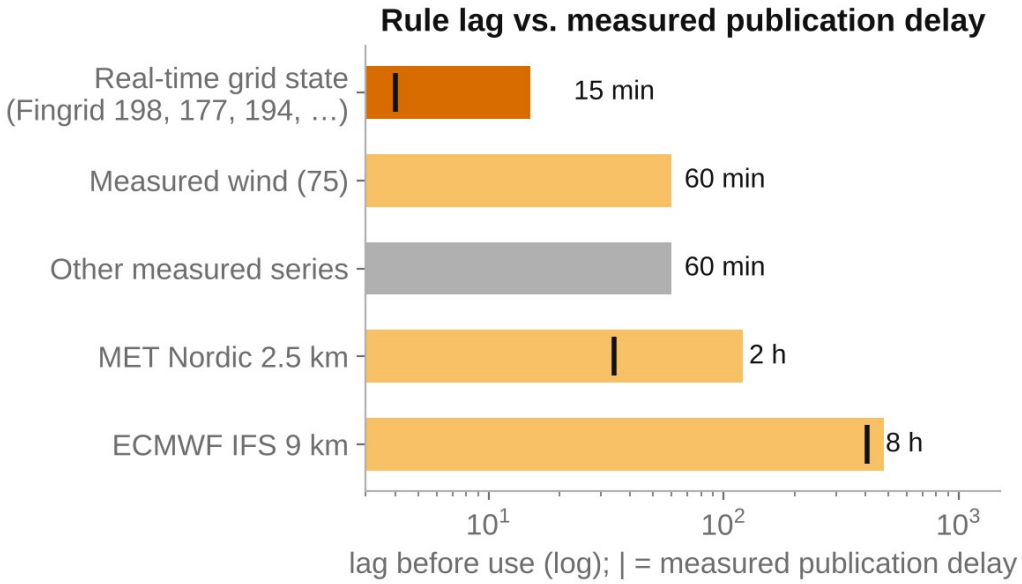
Direction first, then price, all from one distribution

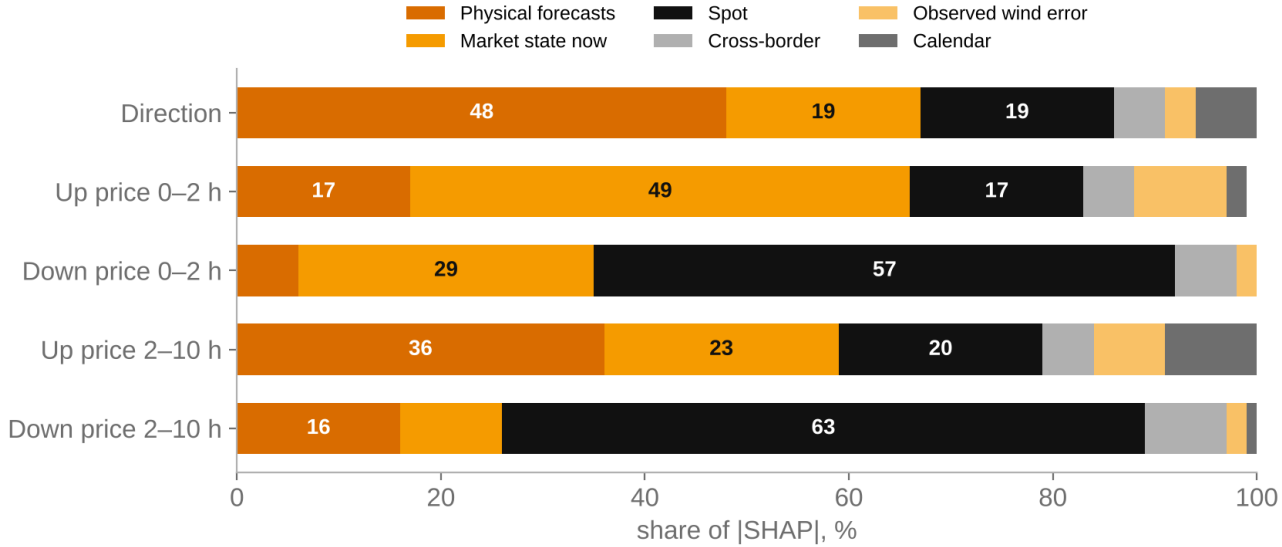


Three scored outputs read from the same distribution: they cannot contradict each other.

LightGBM · separate heads for steps 1–8 and 9–40 · average of 8 members (2 weather sources × 4 variants)

Fed the freshest data the rules allow and proven leak-free





Leakage check

features rebuilt from data with every not-yet-usable row physically deleted; 350 origins, bit-exact match

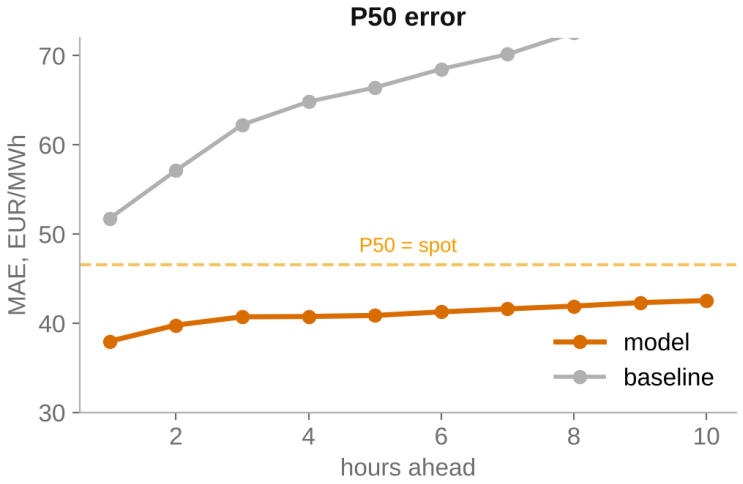
PASS

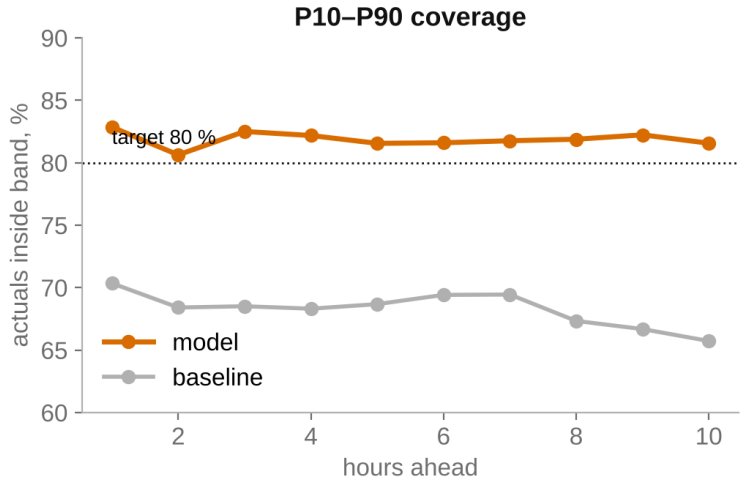
What each part of the model uses

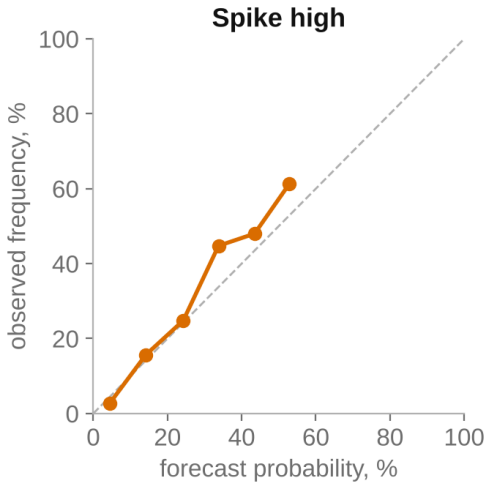
Direction follows the physics; the near-term up price follows the market state; the down price follows spot.

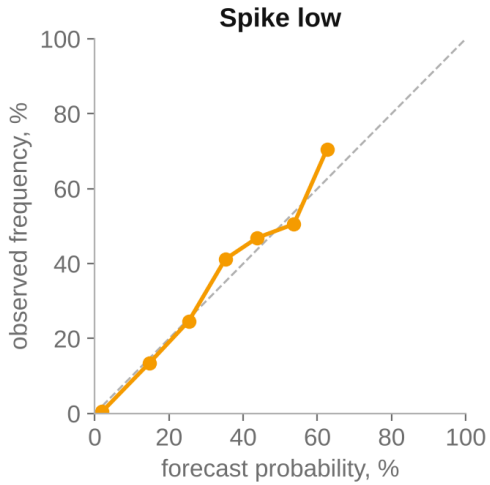
Full SHAP tables and per-row drivers: EVALUATION.md §3.3 on GitHub

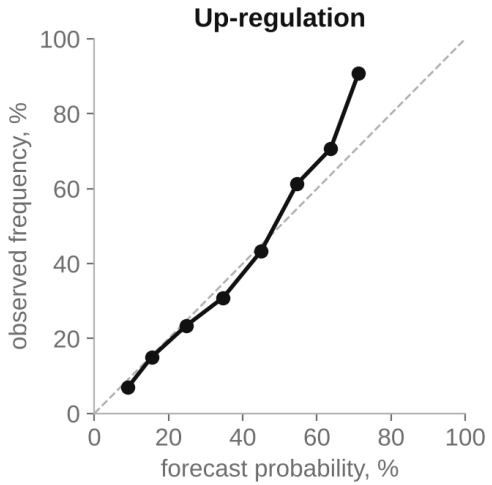
Above the published state of the art and calibrated











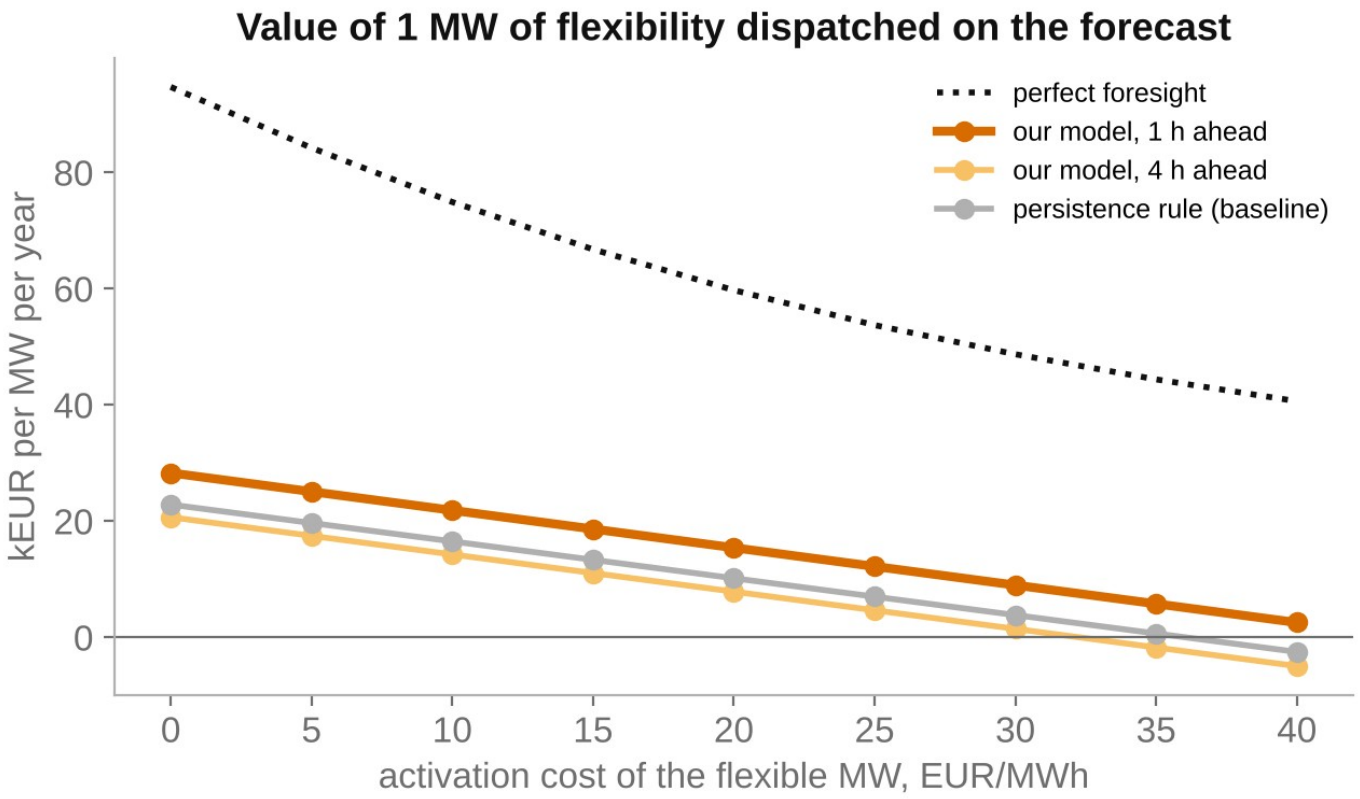
Published 15-min imbalance models: 12–25 % over persistence. Ours: 38 % MAE, 53 % pinball

Belgium 2023 (arXiv 2605.17045), Austria (arXiv 2605.09061). Band coverage 80–83 % in every lead block; baseline 68 %.

Reliability curves on the diagonal

When the model says 40 %, a spike follows about 40 % of the time. No separate post-hoc calibration.

Flexibility dispatched on probabilities earns 50 % more



15 k€

per MW per year with our model

1 h ahead, activation cost 20 €/MWh

10 k€

with the persistence rule

same rule, baseline probabilities

51 k€

with perfect foresight

the headroom still on the table

Rule: raise output or cut load for the quarter-hour when P(spike high) ≥ 0.3; take load when P(spike low) ≥ 0.3. Paid the realised spread, minus the activation cost. Validation outcomes, annualised ×13.

50 MW of electric boilers: about 0.75 M€ a year, 0.25 M€ more than the naive rule.

The operator sees what to check now and how likely it is

Risk of an exceptionally high price has risen for 16:00–18:00: P(spike ↑) 34 %, 3× normal. Drivers: wind forecast down, consumption above forecast, net import falling. Wide band: check the production plan and the market position.

Now and the next 10 h

band, direction, spikes, drivers

In hindsight

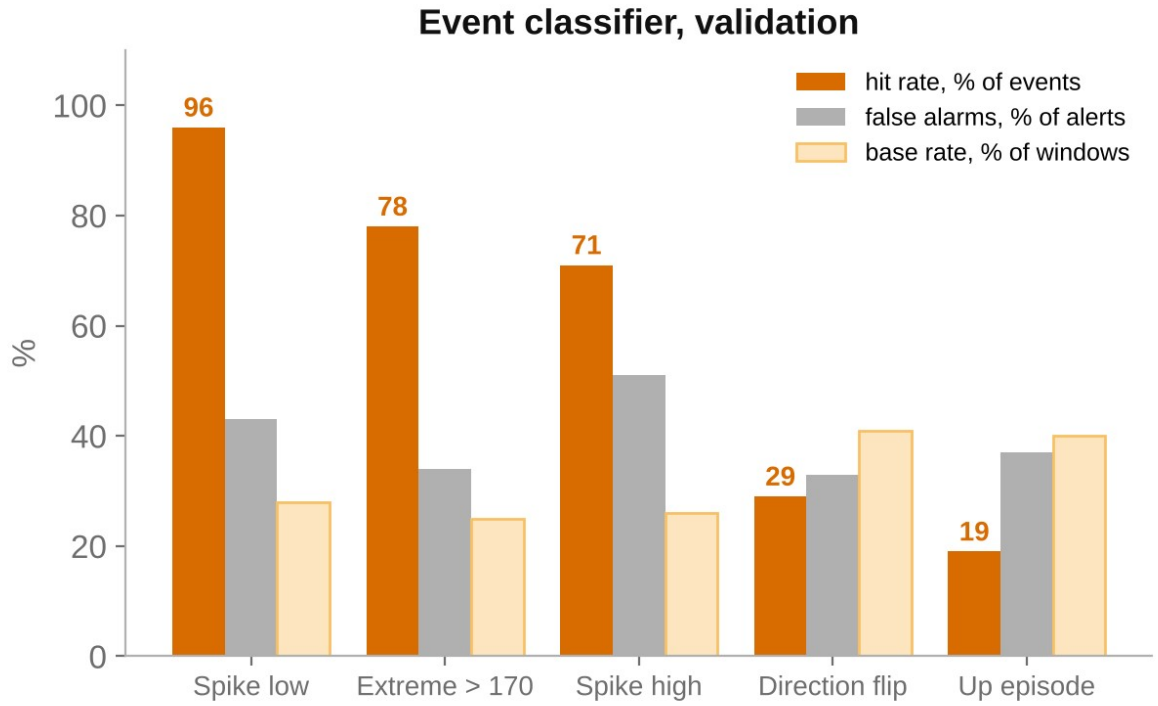
review point movable to any past hour

Alerts

forecast moved, band wide, models disagree

Two roles

trader · CHP + electric boilers



Event classifier: 4 candidate models per event, chosen on validation Brier; Venn–Abers interval on every probability; weak events hidden.

Suggestions to review, not operating instructions.